

Econometrics

Meaning

- Econometrics is the application of statistical and mathematical methods to analyze economic theory with data by using different techniques of estimation and testing of hypotheses relating to economic theories.
- Economic theories are usually expressed in mathematical forms. Statistical methods are adopted in explaining the economic phenomenon in stochastic form that constitutes the econometric methods.
- Econometrics is used to estimate the values of the parameters which are essentially the coefficients of mathematical equations representing economic relationships. The econometric relationships can capture the random behaviour of economic relationships which are not considered in theories in economics.

Variables

- **Dependent Variable-** is the variable that is being observed to see if it changes in response to another variable (independent variable)
- **Independent variable-** is the variable that you manipulate to see if it affects another variable (dependent variable)
- **Intervening/ Mediating variable-** follows the occurrence of the independent variable and precedes the dependent variable
- **Moderating Variable-** is a variable that has a strong contingent effect on the relationship between dependent and independent variables
- **Exogeneous variables-** variables that affect the relationship between dependent and independent variables but have been excluded from the study.

Measurement Scales

Nominal Scale

- Lowest level of measurement.
- Here numbers are assigned for the purpose of identification of objects.
- The assigned number cannot be added, subtracted, divided or multiplied. The only arithmetical operation that can be carried out are the count of each category.

Example: Categories: 1 =State Owned, 2 = Public, 3= Private

Measurement Scales

Ordinal Scale

- An ordinal scale measurement tells whether an object has more or less of characteristics than some other object.
- However, it cannot answer how much more or how much less.
- An ordinal scale tells the relative position of the object and not the difference between the magnitude of the object

Example- Risk Rating: 1 = Very Low Risk, 2 = Low Risk, 3 = Moderate Risk, 4 = High Risk, 5 = Very High Risk

Measurement Scales

Interval Scale

- This scale has all the properties of an ordinal scale, plus the intervals between consecutive values are equal and meaningful. However, it lacks a true zero point. Zero does not signify the absence of the quantity being measured.

Example: Years- 0, 1, 2, 3

Measurement Scales

Ratio Scale

- This is the most informative level of measurement. It has a **true zero point**.
- With ratio data, all mathematical operations—addition, subtraction, multiplication, and division—are valid.

Example: Stock prices

Cross Section Data Analysis

Regression Analysis

- Regression is a primary statistical tool used to analyse cross-sectional data.
- Linear regression model is commonly used to empirically verify the validity of any hypothesis made by the researcher.
- In a linear regression model, relationship is expressed in the form of a linear equation connecting the dependent (explained/response) variable and one or more independent (explanatory) variable
- Simple and Multiple regression analysis

Simple Linear Regression Model

Simple linear regression is a statistical method that allows us to study the relationship between two variables.

$$y = \beta_0 + \beta_1 x + \mu$$

- One variable (x) is the independent variable and other variable (y) is the dependent variable.
- μ is the error term (*factors other than x that influences y*)
- β_0 is the intercept parameter (*value of DV when the IV is Zero*)
- β_1 is the slope (*change in DV for every one-unit change in IV*)

$$\beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

Multiple Linear Regression Model

A linear regression model with more than one explanatory variable is multiple linear regression model

$$y = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n + \mu$$

Ordinary Least Squares (OLS)

- OLS is the method of estimation of linear regression model
- OLS method states that we should select the estimators of the parameters so as to minimize the residual sum of squares.
- By squaring these residuals, OLS ensures that both positive and negative errors are treated equally
- Minimizing this sum means the resulting line is as close as possible to all the data points collectively.

Goodness of Fit

- R^2 (R squared) and Adjusted R^2 as a measure of goodness of fit
- **R-squared**, also known as the **coefficient of determination**, is the most common measure of goodness of fit. It's a value between 0 and 1 that represents the proportion of the variance in the dependent variable that is predictable from the independent variable(s). An R^2 of 1 (or 100%) means the model perfectly explains all the variation in the dependent variable. All data points fall exactly on the regression line.
- A key drawback of R-squared is that it always increases when you add more independent variables to the model, even if those variables aren't statistically significant. **Adjusted R-squared** corrects for this by penalizing the addition of extraneous variables. It's a more reliable measure when comparing models with a different number of independent variables. A negative adjusted R-squared indicates that your model is likely a poor fit for your data and that you should re-evaluate your choice of predictors.

Assumptions of Linear Regression Model/ Diagnostic Tests

Linearity

- Relationship between DV and IV should be linear.
- Scatter plot or rainbow test can be used to check linearity

No Multicollinearity

- Two or more independent variables should not be highly correlated.
- Correlation matrix or Variance Inflation Factor (VIF) can be used to check multicollinearity

Homoscedasticity

- The spread of the residuals is consistent across all values of the independent variables.
- residual plot on scatter diagram or Breusch-Pagan Test can be used to check homoscedasticity

Normality

- Residuals should be normally distributed
- Histogram or Shapiro wilk test

Other types of Regression

- **Logistic Regression**- measures the relationship between dependent and independent variables, when the dependent variable is usually discrete
- **Ridge Regression**- used when there's a strong correlation between independent variables.
- **Lasso Regression**- employed when the linear regression requirements are not met or when the data contains outliers.
- **Stepwise Regression**- gets constructed step by step, resulting in its name. You add or remove predictor variables one at a time depending on the criteria and test the significance for each version. This is different from multiple regression which considers all predictor variables at once.

Time Series Data Analysis

Components of Time Series Data

- **Trends**- Persistent increase or decrease or constant over several years
- **Seasonality**- Pattern that repeats itself at known intervals during a year
- **Cyclical**- Fluctuations that lasts longer than a year and are not of fixed frequency. These fluctuations are usually linked to business or economic cycles
- **Noise**- Residual fluctuations that cannot be explained or predicted

Stationarity

- A time series is stationary if its statistical properties—specifically the mean, variance, and autocorrelation—are all constant over time.
- Most models are built on the assumption that data is stationary.
- Stationarity is required for modelling because, if the data is constantly changing or moving in a very unpredictable way forecasting becomes difficult
- Augmented Dickey-Fuller (ADF) test.
 - Null Hypothesis (H_0): The series is non-stationary
 - Alternative Hypothesis (H_1): The series is stationary.
- We use transformations to make it stationary. The most common technique is differencing.

Autoregressive (AR) and Moving Average (MA)

Moving Average (MA) term (q): This assumes that the current value is dependent on past forecast errors/ short-term shocks.

Autoregressive (AR) term (p): This assumes that the current value is dependent on its own past values (or "lags").

Identifying p and q tells us the structure of the model. Two key diagnostic plots generated from the stationary series:

- Autocorrelation Function (ACF) Plot for MA(q) process
- Partial Autocorrelation Function (PACF) Plot for AR (p) process

Time Series Model- ARIMA

$$y'_t = \overset{\text{intercept}}{c} + \underbrace{\phi_1 y'_{t-1} + \dots + \phi_p y'_{t-p}}_{\text{lagged values}} + \underbrace{\theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}}_{\text{lagged errors}} + \underset{\text{differenced time series}}{\varepsilon_t}$$

- y'_t is the **current change** in the time series being predicted.
- c is the **constant**, representing the series' average trend or drift.
- The **AR part** (ϕ terms) is the effect of **past changes**, capturing the series' momentum.
- The **MA part** (θ terms) is the effect of **past random shocks**, capturing how the series recovers from errors.
- ε_t is the new, **unpredictable random shock** affecting the series in the current period.

Post-estimation Test - Autocorrelation

- The differences between the actual values and the model's predicted values (residuals) should be random noise, with no significant autocorrelation.
- If significant autocorrelation is observed in the residuals, it means the model has failed to capture all the underlying patterns indicating the model can be improved.
- Ljung-Box test is the post estimation test to check for autocorrelation

Hypotheses:

- ✓ *Null Hypothesis : The residuals are random and independent (no autocorrelation).*
- ✓ *Alternative Hypothesis : The residuals are not random (autocorrelation exists).*

Steps

- Declare data to be time series
- Visualize the series
- Test for Stationarity
- Determine AR and MA term
- Estimate ARIMA
- Post estimation Test- Autocorrelation
- Forecasting

Panel Data Analysis

Long vs Wide format

- In long format- variables are placed in columns. Units and time is placed in rows
- In a wide format, units or time are in columns

** For panel data analysis, if data is in wide format, reshape it into long format*

Balanced vs Unbalanced Panel

- A panel dataset is **balanced** when *every* entity (e.g., company) is observed for *every* time period in the dataset.
- A panel dataset is **unbalanced** when there are *missing observations* for at least one entity in at least one time period.

Why is it wrong to run OLS regression?

- Pooling all data together and running one OLS regression on panel data is called “Pooled OLS”
- Why is it wrong?
 - ✓ Ignores panel structure- OLS model assumes all observations are independent
 - ✓ Does not solve the Omitted Variable Bias

Omitted Variable Bias (OVB)

Omitted variable bias is the bias in the OLS estimator that arises when

- Independent variable, X is correlated with the omitted variable.
- The omitted variable is a determinant of the dependent variable Y

As a result, model assigns the effect of the omitted variable (sticky and unobserved) to the independent variable X . This results in biased coefficient, distorted relationship and unreliable conclusion

Primary Types of Panel Data Models

- **Fixed Effects (FE) Model**

- ✓ Units have individual characteristics that may influence the independent and/ or dependent variable.
- ✓ These characteristics need to be controlled
- ✓ Panel data adds a fixed intercept to every unit to control for all sticky omitted variables at once.

- **Random Effects (RE) Model**

- ✓ The variation across entities is assumed to be random and uncorrelated with the independent variables

Fixed Effects Model VS Random Effects Model

- Whenever there is a clear idea that individual characteristics of each entity or group affect the independent variable, use fixed effects. Random effects is used whenever there is reason to believe that individual characteristics have no effect on the independent variable
- The **Hausman-test** tests whether the individual characteristics are correlated with the regressors

Steps

- Declare data to be panel
- Run FE Model & store estimates
- Run RE Model & store estimates
- Run Hausman Test
- Post estimation test- Autocorrelation and heteroscedasticity

References

- Econometrics in Theory and Practice- Panchanan Das
- Basic Econometrics- Damodar N. Gujarati and Dawn C. Porter